

Qatar University
College of Arts and Sciences
Department of Mathematics and Statistics

STAT 481 — R Laboratory 1

First Contact with a Data Matrix — the 47 Swiss Provinces

Out: **Sun, Aug 30, 2026** • Due: **Sun, Sep 6, 2026** by 23:59 (Blackboard)

*Topics: the data array, the mean vector, **S** and **R** (L1–L3)*

Instructor: Prof. Dhafer Malouche • Total: 20 pts

What you submit — two files, nothing else

1. RLab1_Lastname_Firstname_ID.qmd — your Quarto source.
2. RLab1_Lastname_Firstname_ID.pdf — the rendered output of *that* source.

The reproducibility rule. I render your .qmd myself. A report whose source does not render, or that renders to something other than the PDF you submitted, **loses all reproducibility marks** (see the rubric).

The data. `swiss` ships with R — no package to install, no file to download. It records six socio-economic measurements on the **47 French-speaking provinces of Switzerland in about 1888**, at the moment when Swiss fertility began its long fall.

Variable	Meaning	Units
Fertility	standardised common fertility measure	index, 0–100
Agriculture	males employed in agriculture	% of male workforce
Examination	army draftees receiving the highest mark	% of draftees
Education	draftees educated beyond primary school	% of draftees
Catholic	Catholic population	% of population
Infant.Mortality	live births who died before age 1	% of live births

Every column is on a 0–100 scale, which is unusual and convenient: it means you can compare **S** and **R** here *without* the unit problem that dominated `USairpollution` in Lecture 2, and see what *else* the two matrices differ in.

How to work. Open the course project `RProject/STAT481.Rproj`, create your .qmd there, and start it with the setup chunk below. Every task ends in a **number or a figure you must report**: put it in your document and say in one sentence what it means. Install `corrplot` once *in the console*, never inside the .qmd. A plot drawn inside a code chunk lands in the rendered PDF by itself; Lecture 3's `pdf()/dev.off()` pair is not wanted here. The blank space under each task is for planning on paper — your answers go in the .qmd.

```
# ---- setup: everything the report needs, in one place ----
library(corrplot)      # Lecture 2: one picture of a correlation matrix
data(swiss)            # base R; no package needed for the data itself
```

Only the R you have been taught. Everything this lab needs appeared in Lectures 1–3: `dim`, `str`, `head`, `[]` subsetting, `colMeans`, `apply`, `cov`, `cor`, `det`, `plot`, `corrplot`. Four of these you have not typed before: `sum(is.na(swiss))` counts missing cells; `hist(swiss$Fertility)` draws one column; `sqrt()` is the square root; and `abbreviate(colnames(sw), 6)` shortens labels — `colnames()` names the columns the way `rownames()` (Lecture 1) names the rows — on a *copy* `sw <- swiss`, so that the two 6×6 matrices stop wrapping across the page. You do not need — and should not use — anything else.

Collaboration & AI tools. You may discuss ideas with classmates, and you may use AI tools *as a study aid* (checking reasoning, exploring R syntax). You must **write your own report in your own words** and **acknowledge any such use** in a line at the end. You may be asked to explain any part of your submission in class or in office hours; inability to explain your own work forfeits the corresponding marks (syllabus AI policy).

Late work. 10% penalty on submission after the deadline, a further 10% per additional 24 hours (syllabus).

Tasks

Task 1 (Load and describe — the data are real, look at them first; 4 pts)

Load `swiss` and answer, with printed output supporting each answer:

- What are n and p ? What kind of R object is `swiss` (`class()` answers this), and what does one *row* stand for?
- Are there missing values? Report the number.
- Report one thing about these six columns that a table of means and standard deviations does not reveal, and show the output or figure that made you notice it.** There is at least one such thing in this data set, and it will matter in every later task. (A mean and an SD give a column's centre and spread — look at its *shape* instead.)

```
# your code here: dim(), str(), sum(is.na(swiss)), and whatever (c) needs
```

Report: the two numbers n and p , the missing-value count, and your finding in (c) in one sentence.

Task 2 (Compute — the mean vector, **S** and **R**; 6 pts)

- Compute and print $\bar{\mathbf{x}}$, **S** and **R**. State the dimensions of each *before* you run the code, then check with `dim()`. On $\bar{\mathbf{x}}$ that prints NULL: R holds a mean vector as a plain vector of length p , which is not an error.
- Report the pair of variables with the **strongest** correlation, and the value. Say precisely what you mean by “strongest”. (The 1s on the diagonal are not candidates.)
- Verify one entry of **R** a second way.** Take the pair from (b) and recompute its correlation *from **S** alone*, using $r_{jk} = s_{jk} / \sqrt{s_{jj}s_{kk}}$ — pull the three numbers out of `cov(swiss)` by name and

combine them. Store it first — `S <- cov(swiss)` — after which one entry is `S["Fertility", "Catholic"]`; the two variances you need sit on the diagonal. Show that the two agree.

- (d) Compute $|\mathbf{S}|$ and $|\mathbf{R}|$. Which of the two would you quote to a colleague, and why?

Report: $\bar{x}$, $\mathbf{S}$ and $\mathbf{R}$ as printed; the pair and its value; the two determinants.

Task 3 (One figure — you choose it, and you defend it; 5 pts)

This task is deliberately under-specified, and that is the point.

A colleague hands you `swiss` and one question:

“Which of these six variables are telling me the same thing?”

Produce **exactly one figure** that answers it (figures you drew for Task 1 do not count against this). You choose the display, the variables, and whether to show all six or a subset. Then, in **3–4 sentences**, justify the choice using evidence from *your own* output from Task 2 — not from general principles.

Your figure must carry axis or variable labels and a caption saying what to look at — one italic sentence of your own, in the text directly under the figure, is a caption. A figure with none scores zero here, however good it is.

Report: the figure and the justification. A defended choice earns full marks even if I would have chosen otherwise; an undefended one earns none.

Task 4 (Interpret for a non-statistician — in your own words; 5 pts)

Take the strongest correlation you found in Task 2 and explain it, in **3–4 sentences**, to a historian who knows nineteenth-century Switzerland very well and multivariate statistics not at all.

Your answer must contain all three of:

- (a) what the number says about *these provinces* and *these two variables*, in plain language and in the right units;
- (b) one thing the number does **not** say;
- (c) the sentence you would add if the historian asked “so if we had educated more boys, would more of them have passed the army exam?”

Write (c) as that sentence itself, in the words you would say to the historian. **A causal verb applied to these data (“leads to”, “improves”, “drives”) costs the whole of (c).**

Before you submit — tick every line

- ☐ I closed R, reopened the project, and rendered the .qmd from a **fresh session** — it worked.
- ☐ The document **loads its own data** (`data(swiss)`); nothing comes from my console history.
- ☐ No absolute paths (`/Users/...`, `C:\...`).
- ☐ Every number I state appears in printed output somewhere in the report.
- ☐ My one figure has labels and a caption saying what to look at.
- ☐ Both files carry the required name (`RLab1.Lastname.Firstname.ID`).
- ☐ I acknowledged any AI assistance in one line at the end.

Marking rubric (20 pts)

Criterion	Pts	Full marks	Zero
Correctness	8	n , p , $\bar{x}$, S , R , $ \mathbf{R} $ correct; the Task-2(c) verification genuinely goes through S ; every reported number appears in printed output.	Numbers stated that no output supports; or the “verification” is <code>cor()</code> called a second time.
Code quality	4	Readable, commented, one idea per chunk; only functions from the list on page 1; no copy-paste repetition.	Unrunnable or uncommented; or a package the course has not taught, used without a reason.
Interpretation	5	Task 4 names both variables in their units, gives a real limitation and refuses the causal reading; Task 3’s choice is defended with the student’s own Task-2 numbers.	Mechanical restatement of output; an undefended figure choice; or any causal claim from these observational data.
Reproducibility	3	The .qmd renders from a fresh session and produces the submitted PDF; <code>data(swiss)</code> called inside the document.	Does not render, or depends on objects and paths that exist only on the student’s machine.
Total	20		

Grade returned as: correctness ____/8 + code ____/4 + interpretation ____/5 + reproducibility ____/3 = ____/20.